Chapter 1

SUPPLEMENTARY MATERIALS FOR INTRODUCTION TO PYTHON

Installing Anaconda

- What is Anaconda Distribution?
- How it is different from Python Distribution?
- How to install Anaconda?
- conda repository
- Anaconda Navigator
- pip and conda to get new package
- pip and conda commands
- set Virtual
- Integrating Anaconda with Pycharm

Ref: https://www.slideshare.net/slideshow/installing-anaconda-distribution-of-python/86809596

What is Anaconda Distribution

- Anaconda is a freemium open source distribution of the Python and R programminglanguages for large-scale data processing, predictive analytics, and scientific computingthat aims to simplify package management and deployment.Package versions aremanaged by the package management system conda.

- With over 4.5 million users, the open source Anaconda Distribution is the easiest way to do Python data science and machine learning. It includes hundreds of popular data science packages and the conda package and virtual environment manager for Windows, Linux, and MacOS. Conda makes it quick and easy to install, run, and upgrade complex data science and machine learning environments like scikit-learn, TensorFlow, and SciPy.

https://www.anaconda.com/download

Ref: www.ethans.co.in

How it is different from Python Distribution

- Anaconda Data Science Libraries
- Over 1,000 Anaconda-curated and community data science packages
- Develop data science projects using your favorite lDEs, including **Jupyter**, **Spyder** and Rstudio
- Analyze data with scalability and performance with Dask, numpy, pandas and Numba
- Visualize your data with Bokeh, Datashader, Holoviews or Matplotlib
- Create machine learning and deep learning models with Scikit-learn, Tensorflow, h20 and theano

Installing Anaconda

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For installation assistance, refer to Troubleshooting.

Download Distribution by choosing the proper installer for your machine.

Download



Anaconda Installers

Windows

Python 3.12

- 64-Bit Graphical Installer (912.3M)

Mac

Python 3.12

- 64-Bit (Apple silicon) Graphical Installer (704.7M)
- 64-Bit (Apple silicon) Command Line Installer (707.3M)
- 64-Bit (Intel chip) Graphical Installer (734.7M)
- 64-Bit (Intel chip) Command Line Installer (731.2M)

Linux

Python 3.12

- 64-Bit (x86) Installer (1007.9M)
- 64-Bit (AWS Graviton2 / ARM64) Installer (800.6M)
- 64-bit (Linux on IBM Z & LinuxONE) Installer (425.8M)

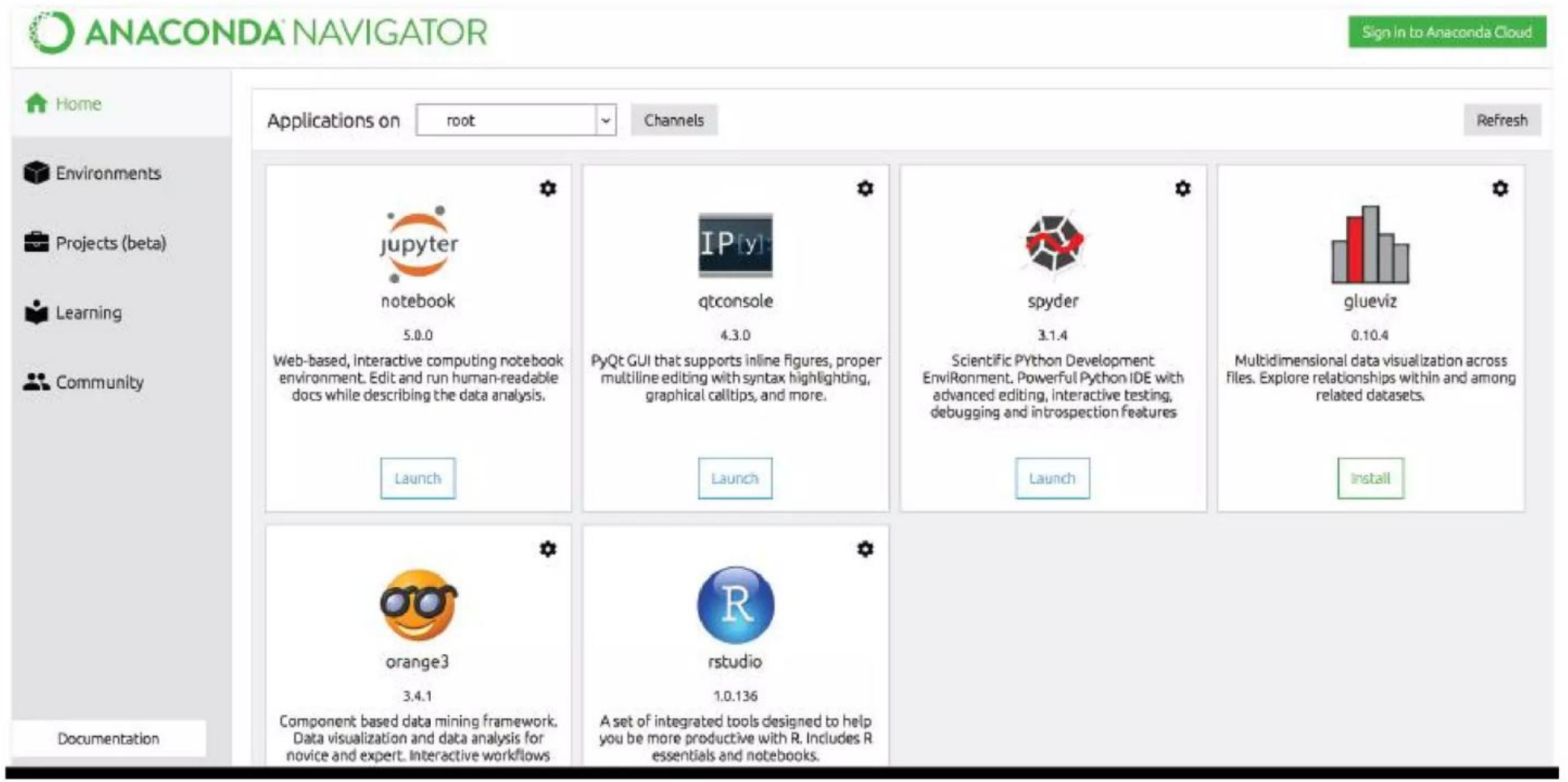
https://www.anaconda.com/download

Conda Repository



- Conda, the Data Science Package & Environment Manager
- Can be used with any programming language or even multi-language projects
- Works across all platforms: Linux, macOS, Windows
- Handles environments natively
- Download conda packages from Anaconda, Anaconda Cloud, Anaconda Enterprise or Conda Forge

Anaconda Navigator



Pip and conda Installer

- pip is a package management system used to install and manage software packages written in Python. Many packages can be found in the Python Package Index (PyPI).
- Python 2.7.9 and later (on the python2 series), and Python 3.4 and later include pip (pip3 for Python 3) by default.
- pip is a recursive acronym that can stand for either "Pip Installs Packages" or "Pip Installs Python".
- pip is quite similar to easy_install. It is a python module that lets you download, build, install, and manage Python packages automatically.
- To install pip, download the get-pip.py file from https://pip.pypa.io/en/latest/installing.html
- To install the package using pip, supply the filename only.

Pip Commands

The most common scenario to install a package using **pip** is:

- pip –h # to display the help page
- pip list # to display the installed packages
- pip list –outdated # to display all outdated package
- pip search xlrd # to search the package in repo
- pip install Package # latest version
- pip install Package==1.0.4 # specific version
- pip install 'Package>=1.0.4' # minimum version requirement
- pip uninstall package # to uninstall the package
- pip install –upgrade # to upgrade the package

Create Virtual Environment

```
C:\Users\Jatin Sir>conda create --name myProject numpy
Solving environment: done

## Package Plan ##

  environment location: C:\Anaconda\envs\myProject

  added / updated specs:
    - numpy
```

Create Virtual Environment

```
                                                    Total:        186.9 MB

The following NEW packages will be INSTALLED:

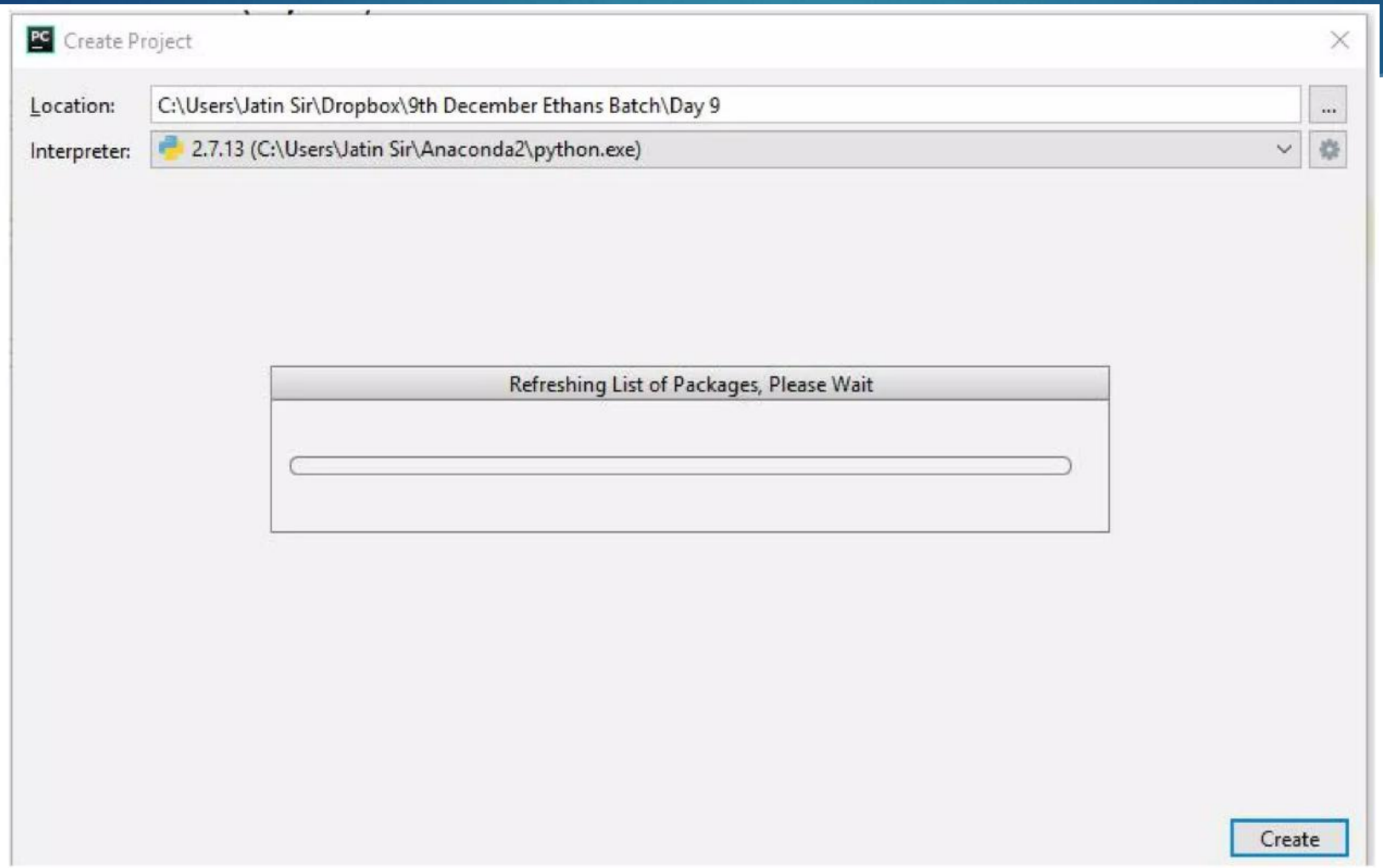
    certifi:          2018.1.18-py36_0
    icc_rt:           2017.0.4-h97af966_0
    intel-openmp:     2018.0.0-hd92c6cd_8
    mkl:              2018.0.1-h2108138_4
    numpy:            1.14.0-py36h4a99626_1
    pip:              9.0.1-py36h226ae91_4
    python:           3.6.4-h6538335_1
    setuptools:       38.4.0-py36_0
    vc:               14-h0510ff6_3
    vs2015_runtime:   14.0.25123-3
    wheel:            0.30.0-py36h6c3ec14_1
    wincertstore:     0.2-py36h7fe50ca_0

Proceed ([y]/n)? y
```

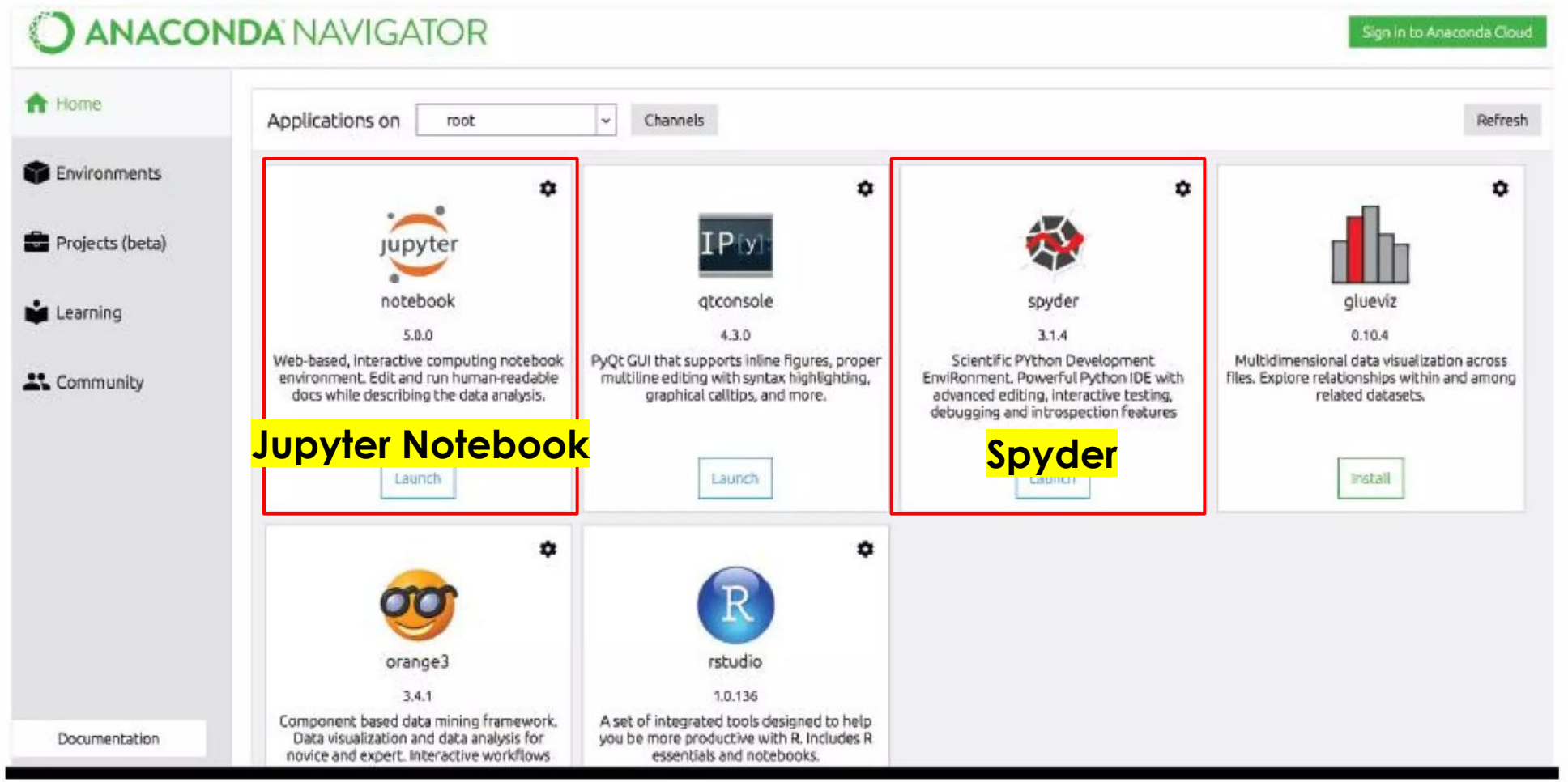
Create Virtual Environment

```
Executing transaction: done
#
# To activate this environment, use:
# > activate myProject
#
# To deactivate an active environment, use:
# > deactivate
#
# * for power-users using bash, you must source
#
```

Integrating Anaconda with PyCharm



Anaconda Navigator



How to open Jupyter Notebook

方法	操作步骤	适用场景
方法一：通过 Anaconda Navigator	1. 在开始菜单中找到并打开 **Anaconda Navigator** 2. 在 Navigator 主界面，找到 **Jupyter Notebook** 图标下的 **“Launch”** 按钮并点击	图形化界面，适合所有用户，特别是初学者。
方法二：通过 Windows 开始菜单	1. 点击 Windows 开始菜单。 2. 找到 **Anaconda3** 文件夹并展开。 3. 直接点击 **Jupyter Notebook** 的快捷方式图标即可。	最快、最直接的方式，一劳永逸。
方法三：通过 Anaconda Prompt (命令行)	1. 在开始菜单中找到并打开 **Anaconda Prompt** 2. 在打开的终端窗口中输入以下命令并回车： `bash jupyter notebook `	更习惯使用命令行的用户，或需要指定特定虚拟环境时